

DCS: 52 process variables and alarm events

1 Perceive

causal window $[21, 41)$, 60 min after onset
a variable is in alarm at 3σ
correlated alarms collapse into one group

2 Score

the 1D-CNN saved for this fold,
loaded from disk, never retrained

3 Reason

retrieval by symptom similarity
over 21 documents, one per class
rules, not a language model

4 Decide

generate (confidence ≥ 0.50)
observe | defer | alert (≥ 10 alarms)
one action per step, each with its guard

macro- F_1
measured
here

recommendation, ranked and cited

Operator: approves, edits or rejects

only after approval

Plant

observe: the window advances 10 samples,
30 min of plant time, at most once